Stroke

BRIEF REPORT

Cumulative Concussion and Odds of Stroke in Former National Football League Players

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BACKGROUND AND PURPOSE: Postmortem and experimental studies indicate a potential association between repeated concussions and stroke risk in older contact sport athletes. We examined the relationship between concussion and stroke history in former National Football League players aged ≥50 years.

METHODS: Former professional football players aged ≥50 years who played ≥1 year in the National Football League were enrolled in the cross-sectional study. Indirect standardization was used to calculate overall and decade-specific standardized prevalence ratios. Logistic regression using Firth's bias reduction method examined the association between lifetime concussion history 0 (n=119; 12.2%), 1 to 2 (n=152; 15.5%), 3 to 5 (n=242; 24.7%), 6 to 9 (201; 20.5%), and 10+(n=265; 27.1%) and stroke. Adjusted odds ratios for stroke were calculated for concussion history groups, age, and coronary artery disease and/or myocardial infarction.

RESULTS: The 979 participants who met inclusion criteria had a mean age of 65.0±9.0 years (range, 50–99). The prevalence of stroke was 3.4% (n=33), significantly lower than expected based on rates of stroke in US men aged 50 and over (standardized prevalence ratio=0.56, Z=−4.56, P<0.001). Greater odds of stroke history were associated with concussion history (10+ versus 0, adjusted odds ratio [95% CI]=5.51 [1.61–28.95]), cardiovascular disease (adjusted odds ratio [95% CI]=2.24 [1.01–4.77]), and age (1-year-increase adjusted odds ratio [95% CI]=1.07 [1.02–1.11]).

CONCLUSIONS: The prevalence of stroke among former National Football League players aged ≥50 years was lower than the general population, with significantly increased risk among those with 10 or more prior concussions. Findings add to the evidence suggesting that traumatic brain injuries are associated with increased risk of stroke. Clinically, management of cardio- and cerebrovascular health may be pertinent to those with a history of multiple prior concussions.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Keywords: aging ■ cerebrovascular accident ■ concussion ■ mild traumatic brain injury ■ stroke ■ traumatic brain injury

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An association between repeated concussions and adverse long-term neurological outcomes in former contact sport athletes, particularly at higher levels of play (ie, collegiate and professional), has been previously reported.[1] Experimental animal studies and human cohorts with more severe traumatic brain injury have suggested that vasculature damage, including cerebral microvascular ischemic changes, occurs as a consequence of head injury, which may elevate long-term outcome risk.[2–4] Limited administrative cohort human studies have also shown associations between traumatic brain injury and risk of subsequent stroke.[4] Despite preliminary evidence, no studies have investigated the association between repeated concussions and risk of stroke in former professional football players. We investigated the association between history of repeated concussion

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Nonstandard Abbreviations and Acronyms

NFL	National Football League

and stroke among former professional football players aged ≥50 years.

METHODS

The data that support the reported findings are available from the corresponding author, upon reasonable request.

Participants

This cross-sectional study followed the Strengthening the Reporting of Observational Studies in Epidemiology guidelines and was approved by Institutional Review Boards at the Medical College of Wisconsin and University of North Carolina at Chapel Hill. Written informed consent was obtained from participants. Participants played at least 1 year of football in the National Football League (NFL), were aged ≥50 years at the time of the survey, and recruited through individual team and organizational communications (Figure I in the Supplemental Material).[5]

A questionnaire involving demographic information, medical history, health behaviors, concussion history/football history, and current function was completed by participants (Table 1). A standard definition of concussion was presented: "a blow to the head that is followed by a variety of symptoms that may include any of the following: headache, dizziness, loss of balance, blurred vision, seeing stars, feeling in a fog, or slowed down, memory problems, poor concentration, nausea, or throwing up. Getting "knocked out" or being unconscious does NOT always occur with a concussion."[5] Self-reported lifetime concussion history was binned into 5 categories: 0 (n=119; 12.2%), 1 to 2 (n=152; 15.5%), 3 to 5 (n=242; 24.7%), 6 to 9 (201; 20.5%), and 10+ (n=265; 27.1%). Participants reported stroke history (yes/no) by indicating that they been told by a physician or other health care professional that they had suffered a stroke.

Statistical Analysis

Prevalence estimates of stroke were calculated by dividing the number of participants with a history of stroke by the total number of participants. Indirect standardization was used to calculate standardized prevalence ratios based on reference data obtained (US men by decade—50–59, 60–69, 70–90, 80+) from the 2018/19 National Health Interview Survey.[6] A priori factors previously reported as being associated with stroke were tested independently as potential covariates: self-reported coronary artery disease and/or myocardial infarction (yes/no); age (continuous); race (identifying as White or as non-White); and cardiovascular disease risk factor burden (number of risk factors present listed in Table 1; range=0–3+).[7] Logistic regression models estimated adjusted odds ratios (aORs) with 95% CIs fit for each potential covariate; factors significantly associated with the odds of stroke were included in the multivariable model. Due to the low prevalence of stroke, Firth's bias reduction method was used to reduce bias of maximum likelihood estimates within the final model.

Table 1. Participant Characteristics

Participants (N=979)	Stroke group (n=33)	No stroke group (n=946)
Demographic, medical, and sport history	**M±SD/n, %**	**M±SD/n, %**
Age	69.5±9.7	64.8±9.0
Race		
White/non-Hispanic	657 (69.5%)	24 (72.7%)
Identified as non-White	289 (30.5%)	9 (27.3%)
Education		
HS graduate, GED, some college	144 (15.2%)	7 (21.2%)
Bachelor's degree	556 (58.8%)	19 (57.6%)
Graduate/professional degree	246 (26.0%)	6 (18.2%)
Concussion history		
0	117 (12.4%)	2 (6.1%)
1–2	148 (15.6%)	4 (12.1%)
3–5	235 (24.8%)	7 (21.2%)
6–9	199 (21.0%)	2 (6.1%)
10+	247 (26.1%)	18 (54.5%)
Lifetime years of football	17.7±4.6	18.5±5.6
Coronary artery disease and/or myocardial infarction	129 (13.6%)	11 (33.3%)
Cardiovascular risk factor burden		
0	256 (27.1%)	5 (15.2%)
1	249 (26.3%)	8 (24.2%)
2	255 (27.0%)	13 (39.4%)
3+	186 (19.6%)	7 (21.3%)
Cardiovascular risk factors		
Hypertension	461 (48.7%)	20 (60.6%)
Hyperlipidemia	329 (34.8%)	11 (33.3%)
Cigarette use	187 (19.8%)	7 (21.2%)
Obstructive sleep apnea	288 (30.4%)	13 (39.4%)
Type-II diabetes	79 (8.4%)	3 (9.1%)
Chronic obstructive pulmonary disease	18 (1.9%)	3 (9.1%)

RESULTS

Stroke prevalence (n=33; PR [95% CI]=3.4% [2.2%–4.5%]) among former NFL players was significantly lower than expected on the basis of rates of stroke in the general population of US men aged 50 and over (expected events=59.2; standardized prevalence ratio=0.56; Z=−4.56, $P<0.001$). PRs by decade are: 50 to 59 (PR [95% CI]=1.9% [0.4%–3.4%]), 60 to 69 (PR [95% CI]=3.1% [1.3%–4.9%]), 70 to 79 (PR [95% CI]=3.4% [2.2%–4.5%]), 80+ (PR [95% CI]=12.3% [3.2%–21.4%]; Figure II in the Supplemental Material). Of the potential covariates, coronary artery disease and/or myocardial infarction (OR [95% CI]=3.25 [1.53–6.91]), and age (1-year-increase OR [95% CI]=1.06 [1.02–1.10]) were significantly associated with stroke and included in the multivariable model. Race and cardiovascular disease risk factors were not significantly

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associated with higher odds of stroke. Adjusting for covariates, those with a history of 10 or more prior concussions had 5× the odds of stroke compared with those with no prior concussions (aOR [95% CI]=5.51 [1.61–28.95]; Table 2; Figure). Higher odds of stroke was also associated with older age (1-year increase aOR [95% CI]=1.07 [1.02–1.11]) and history of coronary artery disease and/or myocardial infarction (aOR [95% CI]=2.24 [1.01–4.77]). Interactions effects for variables within the model were not observed.

DISCUSSION

Stroke prevalence in this sample of former NFL players was significantly lower than the general male population of comparable age. Findings suggest that the subset of former professional football players with 10 or more concussions, which accounted for one-quarter of our sample, had a significantly greater probability of stroke compared with those who sustained fewer concussions. This supports the growing evidence that traumatic brain injury, and repeated traumatic brain injury specifically, is an independent risk factor for stroke. The absence of a dose-response association may be due to lower statistical power (ie, rare event), or reflect a threshold effect where cumulative concussion places individuals at risk for stroke at a critical magnitude. Management of cardio- and cerebrovascular health is particularly important for clinicians caring for those with a history of multiple concussions.

Previous studies on former NFL players have reported the presence of white matter abnormalities revealed by neuroimaging.[8] These changes in white matter possibly indicate more advanced macroscopic cerebrovascular changes and are predictive of stroke in multiple prospective community cohorts,[7,9] representing one potential mechanism that may increase risk for stroke among individuals with greater history of concussion.

A limitation of this study includes the cross-sectional nature of the data, limiting causal inferences.

Table 2. Logistic Regression Model of Stroke*

Effect	Adjusted odds ratio (95% CI)†
Concussion history‡	
1–2	1.59 (0.34–9.32)
3–5	1.82 (0.47–9.99)
6–9	0.81 (0.12–5.37)
10+	5.51 (1.61–28.95)
Age (1-y increase)	1.07 (1.02–1.11)
Cardiovascular disease	2.24 (1.01–4.77)

*Logistic regression using the Firth method (ie, bias reduction of maximum likelihood estimates for rare-events).

†Profile-likelihood 95% CI (lower level, higher level).

‡Zero prior concussion serves as referent group.

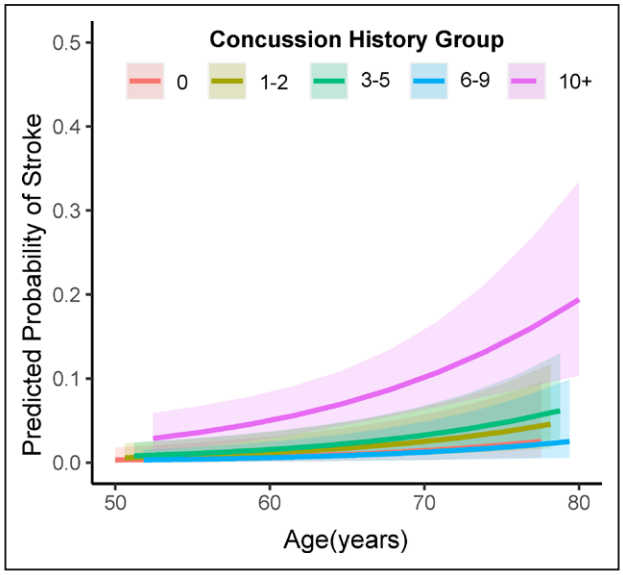

Figure. Predicted probability of stroke history. Distribution according to age of predicted probability and 95% CIs of stroke history reporting in concussion history groups among participants.

Medical conditions, including concussion history, were self-reported. Though variable consistency in concussion reporting can occur, we would not expect this to differ across those with and without a history of stroke. Lower prevalence estimates of stroke in this sample may reflect response bias (ie, severe stroke less likely to respond), or the cumulative benefit of higher-level physical activity/exercise throughout early decades of life in the sample. The generalizability of the findings to those who participated in lower levels of contact sports is unknown. Given the limited number of events, we were not able to confidently determine the association between individual vascular risk factors and the prevalence of stroke. The lower response rate of the survey is also a limitation of the data. Results suggest lower prevalence of stroke in this sample of former NFL players, with significantly higher risk of stroke among former players with 10 or more prior concussions, independent of age, and cardiovascular disease.

ARTICLE INFORMATION

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Supplemental Materials

Supplemental Figures I and II
STROBE checklist

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